

An Agenda of Mathematical Blind Spots

What this is. An expansion of Part VI of the *Landscape* survey into a standalone research agenda, written *after* the six per-approach deep dives (Groups 1–6) and drawing on them throughout. The audience is a mathematician looking for places where the mechanistic interpretability (MI) literature leans on a structure it never makes precise — and where making it precise is a tractable contribution rather than a slogan. It assumes the conventions of the *ML Concepts* primer (residual stream / feature space V ; QK as a bilinear form $W_{QK} : V \rightarrow V^*$ vs. OV as an endomorphism $W_{OV} : V \rightarrow V$; adjoint W^* vs. dual map $W^\vee$; the $GL(V)$ gauge of the stream; the Gram matrix $G = D^*D$).

How to read this

Seven problems, ordered so that the most load-bearing comes first. They are not independent: a single fault line — *the field computes in a metric it never chose* — runs through the first three, and a second — *the field never says what “correct” means* — through the next two. The last two (Lie structure, harmonic analysis) are higher-risk lenses that might unify the rest. For each problem I give the precise setup, what is already known, what is open, and a concrete first step that a mathematician could take without waiting for any ML infrastructure. Throughout, “leverage” means: if solved, how many of the field’s quantities become well-defined or how many empirical findings get explained.

A one-line map of the dependency structure, expanded at the end: **P1** (the metric/gauge problem) is the spine; **P2** and **P3** are what P1 buys you (a meaningful Gram matrix; a meaningful curvature); **P4–P5** are a parallel spine about approximation and correctness; **P6–P7** are the speculative unifiers.

P1. A gauge theory of representations

The problem. The residual stream carries no privileged basis: conjugating the network by any $M \in GL(V)$ — replacing each read map $W_{\text{in}} \mapsto W_{\text{in}}M^{-1}$ and each write map $W_{\text{out}} \mapsto MW_{\text{out}}$ — leaves the function bit-identical [1, 7]. Yet the field’s most-used quantities are computed in the standard basis: cosine similarity between feature directions, the ℓ_2 reconstruction loss of a sparse autoencoder [6, 11], the off-diagonal of the Gram matrix that defines interference [3]. All three are *gauge-dependent*: they change under M . The field works in a metric it never chose, and exactly one paper notices [10].

What is known. The invariance principle can be stated cleanly. A scalar is gauge-invariant iff it is built only from the model’s own maps and the natural pairing of a space with its dual. On the invariant side:

- the embedding–unembedding pairing $\langle \lambda(x), \gamma(y) \rangle$ the softmax consumes, legitimate because the embedding space is the dual of the unembedding space, $\Lambda \cong \bar{\Gamma}^*$ [10];
- the attention score $\langle W_{QK}x_k, x_q \rangle$, a learned bilinear form whose query/key sides transform contragrediently;
- a linear probe $\langle w, x \rangle$ with $w \in V^*$ a covector;

- the inter-block *virtual weight* $W_{\text{in}}^B W_{\text{out}}^A$, and with it the metric-free relation “ A never affects B ”
 $\iff \text{im}(W_{\text{out}}^A) \subseteq \ker(W_{\text{in}}^B) \iff W_{\text{in}}^B W_{\text{out}}^A = 0$.

On the dependent side: anything that pairs two vectors of the *same* space through the standard inner product, i.e. silently inserts the identity $G = I$ as a metric. The one principled metric in the corpus is the **causal inner product** $\langle \bar{\gamma}, \bar{\gamma}' \rangle_C = \bar{\gamma}^\top \text{Cov}(\gamma)^{-1} \bar{\gamma}'$, i.e. the map $g_C = \text{Cov}(\gamma)^{-1} : \bar{\Gamma} \xrightarrow{\sim} \bar{\Gamma}^*$ [10]; but it is derived only for the *output* space, and it is *not unique* — causal orthogonality fixes $d(d-1)/2$ of the $d(d+1)/2$ degrees of freedom of a symmetric form, leaving $\approx d$ undetermined (the paper chooses them by fiat).

What is open.

- *The interior.* There is no canonical, or even derived, metric for the residual stream *between* embedding and unembedding — exactly the space every SAE and probe operates in. What principle (data covariance? a transport from the output metric backward through the frozen-linear model? invariance under the training dynamics?) should pick it?
- *A gauge-invariant rewrite of the toolkit.* Restate superposition capacity, feature splitting, probing, and SAE reconstruction in $GL(V)$ -invariant terms. Concretely: *what should an SAE minimise if not the gauge-dependent ℓ_2 loss?* A reconstruction error in a derived metric g , $\mathbb{E} \|x - \hat{x}\|_g^2$, is the obvious candidate; the content is in deriving g .
- *The residual d degrees of freedom.* Is the non-uniqueness of the causal inner product a genuine indeterminacy of the model, or an artifact of the binary-concept setup?

Normalization as Lie-subgroup reduction. A clean, checkable sub-result: the architecture’s normalization choice *selects* the residual gauge group, and hence its Lie algebra. With no normalization the gauge is all of $GL(V)$ (algebra $\mathfrak{gl}(V)$). RMSNorm references only $\|x\|$, so it survives only $M \in O(V)$ (algebra $\mathfrak{so}(V)$, the antisymmetric maps). LayerNorm additionally references the mean, i.e. the line $\mathbb{R}\mathbf{1}$, reducing the gauge to $\text{Stab}_{O(V)}(\mathbf{1}) \cong O(d-1)$. Read this way, Elhage et al. [7]’s RMSNorm-vs-LayerNorm experiments are, unknowingly, a study of *which Lie subgroup the architecture leaves unbroken*; their finding that Adam’s per-coordinate normalization is what actually breaks the basis says the *optimizer*, not the architecture, performs the final reduction $O(V) \rightarrow \{e\}$.

First steps (metric-free, no ML stack needed). (i) Write the gauge action on every named object in the corpus and tabulate invariant vs. dependent. (ii) Prove the SAE objective is $O(V)$ -but not $GL(V)$ -invariant, and characterise the orbit of “equivalent” dictionaries. (iii) Propose and analyse one derived interior metric and re-derive the incoherence condition $G \approx I$ in it.

P2. The interference Gram matrix as a first-class object

The problem. Group 2’s deep dive shows that essentially every quantitative claim about superposition is, under the hood, a statement about one matrix: the Gram matrix $G = D^* D$ of the feature dictionary $D : \mathbb{R}^F \rightarrow V$, $G_{ij} = \langle d_i, d_j \rangle$ [3, 13]. Its diagonal is 1 (unit features), its off-diagonal is *interference*, and “near-orthogonality” means $G \approx I$. The literature treats G implicitly; it deserves to be the explicit object.

What is known. The forward geometry of G is partly mapped: optimal feature configurations solve a generalized Thomson problem and land on uniform polytopes / tegum products [3]; the existence of superposition is the metric-free statement that D is non-injective; the cost is the metric-dependent operator norm $\|G - I\|$. Compressed sensing supplies a capacity bound through the coherence $\max_{i \neq j} |\langle d_i, d_j \rangle|$ and the restricted-isometry constant.

What is open.

- *Spectrum and block structure.* What is the eigenvalue distribution of G for trained models, and does **feature splitting** [6, 11] correspond to a hierarchical / nested-block structure of G as F grows? A renormalization-style account of splitting as block-refinement of G would be new.
- *From representation interference to weight interference.* The off-diagonal of G (representation-level) induces spurious “interference weights” between features at the computational level [13]. Make the map $G \mapsto$ (interference weights) precise — it is presumably a quadratic/bilinear image of $G - I$ through the surrounding weights — and bound the latter by the former.
- *Sharp capacity.* Tighten the sparsity-dimension- $\|G - I\|$ trade-off beyond the order-of-magnitude compressed-sensing bound; the toy models hint at phase transitions (the “sticky” dimensionalities $1/2, 2/3, 3/4, \dots$) that a packing-theoretic analysis might pin down.

The whole problem inherits the metric-dependence of P1: $G = D^*D$ uses the adjoint D^* , hence a metric, so “how close to I ” is only meaningful once P1 fixes the inner product.

First step. Take the trained one-layer SAE dictionaries that are public-style replicable from Bricken et al. [6] (or the toy models, which are fully specified) and study G as a random-matrix / packing object: spectrum, coherence, and the geometry of its near-null structure under feature splitting.

P3. A geometry of feature manifolds

The problem. The dominant working definition — a feature is a *direction*, a ray in V — is contradicted, for continuous or ordered quantities, by Gurnee et al. [15]: a scalar count is carried by a curved 1-dimensional *manifold* (a rippling helix) in a low-dimensional subspace, and the model exploits the curvature. The linear-representation hypothesis (LRH) and the manifold picture are, as the *Landscape* survey argues, mathematically incompatible accounts of what a feature *is*, and nobody has reconciled them.

What is known. For the counting task the manifold is approximately an orbit of a one-parameter subgroup (a helix is $\exp(tX)$ for a fixed screw generator X), and an attention head compares two counts by a W_{QK} that rotates/twists one manifold onto another — a near-isometry $\exp(\theta\Omega)$, $\Omega \in \mathfrak{so}$. The “ringing” of the embedding is (provably, in a toy model) the near-optimal low-rank projection of a circle.

What is open.

- *The right invariants.* What is the correct invariant description of a count/colour/date manifold — intrinsic dimension, curvature, holonomy — and in *which* metric (this is P1 again: curvature is metric-dependent)? Is the “feature manifold” best seen as a homogeneous space G/H for a group G acting on V ?
- *Is LRH a special case?* Conjecture: the LRH (feature = direction) is the *flat* / *categorical* degeneration of a manifold theory — exact for unordered categorical concepts (where the manifold is a discrete set of rays) and false for ordered/continuous ones (where curvature is load-bearing because it lets a linear W_{QK} implement a thresholded comparison that a ray cannot). Formalise and test the dividing line.
- *Attention as a near-isometry.* Make precise the sense in which a head acts as a near-isometry between feature manifolds, and what “near” costs.

- *Unsupervised discovery.* The manifold in Gurnee et al. [15] was found by hand. Is there a method (a manifold-learning analogue of dictionary learning) that surfaces such structure automatically?

First step. Pin down, for the single worked example (the count helix), the cleanest invariant statement of its geometry, and prove the ray-vs-curve dichotomy for the comparison operation: that a 1-D ray representation forces a monotone (un-thresholdable) comparison while a curved ≥ 2 -D one admits a genuine rotation-based threshold.

P4. Error semantics of the three “linearities”

The problem. The corpus uses the word “linear” for three operations with different error guarantees, and conflating them is where overclaiming creeps in (Group 4’s deep dive makes this its spine):

1. **Exact, by freezing the nonlinearity** [1]: with attention frozen, the forward pass is *honestly* linear in tokens — zero error, but valid only on the input it was frozen on, and it discards the derivative of the attention pattern (the QK routing).
2. **Local linearization** [12]: a Taylor/Jacobian expansion (the “local replacement model” with stop-gradients and error nodes), exact at one prompt and only first-order away from it; errors compound across layers.
3. **Learned sparse approximation** (SAEs/transcoders): a trained low-error fit that is neither exact nor a derivative, and may be *mechanistically unfaithful* [14].

What is open.

- *Quantify each.* For (2), bound the divergence of the local replacement model as a function of distance from the operating point and depth — the field observes compounding error empirically but has no bound. The relevant object is a product of layer Jacobians; its conditioning controls the blow-up.
- *The influence operator.* Attribution-graph node influence is the Neumann series $B = (I - A)^{-1} - I$ for the (input-normalized) adjacency A [12]. When does it converge, and what does its conditioning / spectral radius say about which circuits are robustly attributable?
- *The combinatorics of QK attribution.* Unfreezing the QK form [16, 17] expands the score into feature–feature pairings; the per-head matrix is “typically approximately low-rank.” Make “approximately low-rank” a theorem and exploit it to compress the otherwise quadratic attribution.

First step. Write the local replacement model as a composition of affine pieces with remainder, and state the first nontrivial bound on its off-prompt error in terms of the layer Jacobians’ norms — the quantity Group 4 needs and does not have.

P5. A quantitative theory of mechanistic faithfulness

The problem. Group 6’s cautionary result [14]: a transcoder can be monosemantic, sparse, and low-MSE yet implement a *different* algorithm that generalises differently off-distribution. “Mechanistic faithfulness” is currently a spectrum without a metric, and the field has no formal definition of the two objects it most relies on — “feature” and “circuit.”

What is known. There is a clear qualitative hierarchy of how strongly a surrogate g can match

a model component f :

$$\underbrace{g \approx f \text{ on-distribution}}_{\text{function match (weakest)}} \subsetneq \underbrace{Dg \approx Df}_{\text{Jacobian match}} \subsetneq \underbrace{\text{same step-by-step trajectory}}_{\text{mechanism match (strongest)}}.$$

Anthropic Interpretability Team [14] proposes **Jacobian matching** (penalise $\|J_{\text{MLP}} - J_{\text{transcoder}}\|_F^2$) as a partial fix, and notes the subtlety that superposition means one should *not* match the Jacobian exactly (a feature spread over k neurons contributes a tail of small singular values that is a superposition artifact). Wang et al. [5] already separates *faithfulness* (matching behaviour) from *completeness* (matching it under all knockouts) — and shows faithfulness $\nRightarrow$ completeness.

What is open.

- *An off-distribution generalization bound.* Make precise: function-match controls error on the training distribution; Jacobian-match controls it to first order in a neighbourhood; trajectory-match controls it globally. State the theorem and the constants. This directly formalises “generalises differently off-distribution.”
- *What to match, given superposition.* Determine the right *partial* Jacobian-match target (which singular subspace to match, which tail to ignore) as a function of the superposition structure G of P2.
- *Define “feature” and “circuit.”* Is a feature a direction (P1), a manifold (P3), or an equivalence class of either under the gauge? Is a circuit a subgraph that is faithful, complete, *and* minimal [5], and is that triple even well-posed given redundancy (Backup heads)?
- *What is an “axiom” of a circuit?* Olah [9] dreams that for small circuits “statements about their behaviour become questions of mathematical reasoning.” What, precisely, would such a statement be, and what are its inference rules?

First step. Prove the hierarchy bound on a fully-specified toy (the $y = \text{abs}(x)$ transcoder of Anthropic Interpretability Team [14]): exhibit two surrogates that function-match but differ in Jacobian, and bound their off-distribution divergence by the Jacobian gap.

P6. Latent Lie structure

The problem. The corpus never invokes Lie algebras or commutators, yet Lie *groups* are pervasive — as gauge symmetries (P1) and as the “addition-as-rotation” geometry of algorithmic tasks — and the only reason brackets are absent is that the field’s additive core is *abelian* by design (the residual stream adds). Non-commutativity enters exactly through (a) composition of maps and (b) group actions. Naming this structure is a lens, not yet a theorem.

What is known / proposable.

- *Gauge Lie algebras (from P1).* $\mathfrak{gl}(V) \supset \mathfrak{so}(V) \supset \mathfrak{so}(V) \cap \mathbf{1}^\perp$ as the normalization is tightened; the infinitesimal side of the gauge theory.
- *Compact tori in grokking.* Nanda et al. [8] represents $\mathbb{Z}/n\mathbb{Z}$ on circles $a \mapsto (\cos \omega_k a, \sin \omega_k a)$ — a homomorphism into the abelian Lie group $U(1)^k$, with addition the group law. Abelian, so every bracket vanishes (which is *why* only Fourier characters, never a commutator, are needed).
- *Non-abelian rotations in counting.* The count manifold (P3) is a one-parameter-subgroup orbit, and the comparison head is $\exp(\theta\Omega)$, $\Omega \in \mathfrak{so}$; in the observed ~ 6 -dimensional subspace the relevant rotations need not commute, so genuine $\mathfrak{so}(6)$ with nonzero brackets is present and unremarked.

- *A gauge-invariant interaction measure.* OV circuits are endomorphisms, so under the gauge they conjugate, $W_{OV} \mapsto MW_{OV}M^{-1}$, and therefore

$$[W_{OV}^A, W_{OV}^B] \mapsto M[W_{OV}^A, W_{OV}^B]M^{-1}.$$

The commutator’s rank, spectrum, and vanishing are *gauge-invariant*. So “do two heads’ value-transformations commute?” is a canonical, metric-free question — unlike the cosine of two feature directions — and $\|[W_{OV}^A, W_{OV}^B]\|$ -class is the kind of invariant interaction measure P1 asks for. (The QK form $V \rightarrow V^*$ has no intrinsic commutator; forming one needs a metric, so QK interaction measures are *gauge-dependent* — itself a consistent echo of the OV/QK type distinction.)

What is open. Whether the commutator measure tracks anything the field cares about (reorderability of layers in the path expansion; whether two circuits can be merged); whether the non-abelian rotation structure in counting generalises to other “manipulate a quantity” tasks; whether the gauge Lie algebra has Noether-style conserved quantities under gradient flow (the rescaling symmetry of Elhage et al. [3] and Elhage et al. [2] is a $\mathbb{R}^\times$ subgroup whose conserved charge is known).

First step. Compute $[W_{OV}^A, W_{OV}^B]$ for the head pairs of a known circuit (induction; the IOI Name Movers [4, 5]) and test whether its (gauge-invariant) rank/spectrum distinguishes interacting from non-interacting pairs better than cosine similarity does.

P7. Harmonic analysis as a possible unifier

The problem. Three different branches of mathematics describe “the geometry of representations” in this corpus and the field has not unified them: finite-group representation theory (the $\mathbb{Z}/n\mathbb{Z}$ characters of grokking [8]), continuous differential geometry (the curved manifolds of [15]), and discrete packing / polytopes (the Thomson geometry of superposition [3]). The tantalising hint is that *periodicity* / *harmonic structure* appears independently in the first two — group characters in grokking, the “ringing”/Gibbs geometry of the count manifold — and that a circulant Gram matrix (P2) has a Fourier eigenbasis.

What is open / the conjecture. Is there a single framework — Fourier analysis on the relevant group, spectral geometry of the relevant manifold, and the eigenstructure of the relevant (circulant or near-circulant) Gram matrix as facets of one harmonic-analytic object — under which the polytope, character, and manifold pictures are special cases? This is the most speculative item and possibly the most fruitful: if the “natural” coordinates of a representation are harmonic (eigenfunctions of some symmetry), then P1’s metric, P3’s curvature, and P2’s Gram spectrum are all read off the same decomposition.

First step. Test the narrow version: for a task with a known symmetry group Γ (cyclic, dihedral, symmetric), does the trained representation decompose into isotypic components of Γ , and is the learned metric (P1) the one that makes that decomposition orthogonal? Grokking already gives the cyclic case; the dihedral/symmetric cases are clean next experiments.

How the problems interlock

The agenda has two spines and two unifiers.

The metric spine (P1→P2, P3). P1 is upstream of almost everything: the Gram matrix of P2 and the curvature of P3 are both metric-dependent, so “how close to orthogonal” and “how curved”

are only well-posed once P1 fixes (or derives) the inner product. Solve P1 and P2/P3 become quantitative; ignore it and they remain basis-dependent folklore. This is why P1 is ranked first despite being the most foundational rather than the most immediately publishable.

The approximation/correctness spine ($P_4 \rightarrow P_5$). P4 is about the error of the *methods* (exact vs. local-linear vs. learned), P5 about the error of the *interpretations* (function vs. Jacobian vs. mechanism match). They meet at the Jacobian: P4’s local-linearization error and P5’s Jacobian-matching target are the same derivative, viewed from the method side and the epistemic side respectively.

The unifiers (P_6, P_7). P6 (Lie structure) is largely a reorganisation of P1 (gauge groups) and P3 (rotation/orbit geometry) into a single algebraic language, plus one genuinely new tool (the gauge-invariant OV-commutator). P7 (harmonic analysis) is the bet that P1’s metric, P2’s spectrum, and P3’s curvature are facets of one spectral object; it is the riskiest and would subsume the most.

Where a mathematician has the clearest edge. P1, P5, and P6 are the items that are mostly mathematics — gauge invariants, approximation bounds, Lie-algebraic invariants — and need the least ML infrastructure to make progress on: each has a “first step” above that is a self-contained lemma on a fully-specified toy model. P2 and P3 reward someone who will also touch trained weights. P7 rewards someone willing to gamble on a unification that may not exist.

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